

Lesson 1: Introduction to Version Control & Git Setup

Welcome to **Practical Git & GitHub: Team Collaboration Essentials**. In this first lesson, we will lay the foundation for how software teams track changes, collaborate without overwriting each other's work, and set up your local development environment.

1. Why Do We Need Version Control?

Imagine working on a project where every update requires saving a new file like `project_v1.py`, `project_final.py`, and `project_final_v2_FINAL.py`. In a team setting, this approach quickly causes lost work, confusion, and broken code.

Version Control Systems (VCS) solve this by acting as a detailed logbook for your code. Git allows you to:

- **Track History:** Revert to any previous state of your project if something breaks.
- **Collaborate Safely:** Work on new features in isolation without disrupting the main codebase.
- **Audit Changes:** See exactly *who* made a change, *when* they made it, and *why*.

2. The Three States of Git

To use Git effectively, you need to understand how it manages files locally. Git moves your project files through three main areas:

[Working Directory] ---> [Staging Area] ---> [Local Repository]

(Making changes) (git add) (git commit)

1. **Working Directory:** The actual files you are currently editing on your computer.
2. **Staging Area (Index):** A draft area where you select and organize the specific changes you want to save.
3. **Local Repository:** The permanent history database on your machine where Git stores committed snapshots of your project.

3. Configuring Your Git Identity

Before making your first commit, Git needs to know who you are. This information is attached to every change you save.

Open your terminal or command line and run the following configuration commands:

Bash

Set your display name

```
git config --global user.name "Your Name"
```

Set your email address (use the email associated with your GitHub/GitLab account)

```
git config --global user.email "your.email@company.com"
```

Verify your settings

```
git config --list
```

4. Initializing Your First Repository & Making a Commit

Let's walk through creating your first tracked project step-by-step.

Step A: Initialize the Repository

Navigate to a project folder and transform it into a Git repository:

Bash

```
mkdir my-first-repo
```

```
cd my-first-repo
```

```
git init
```

This creates a hidden .git folder that tracks all future changes.

Step B: Create a File and Check Status

Create a simple text file and check how Git views it:

Bash

```
echo "# My Project" > README.md
```

```
git status
```

*Git will show README.md under **Untracked files**.*

Step C: Stage and Commit the File

Move the file to the Staging Area and permanently record the snapshot:

Bash

```
# Stage the file
```

```
git add README.md
```

```
# Commit the staged file with a clear message
```

```
git commit -m "Initial commit: Add README file"
```

Step D: View Project History

Check your commit history to verify your save point:

Bash

```
git log
```